Department of Computer Science and Engineering,
Anna University, Chennai- 600025
CS23005 – Business Analytics Lab
Practical
July – November 2026
Year/Sem : IV / VII

Exercise: 02	Descriptive Data Analysis using excel	10 – Jul - 2026

Observation (5 Marks)

1. Compare Mean, Median, and Mode with examples and discuss their applications.
2. Analyze the importance of Standard Deviation in measuring data consistency.
3. Explain Correlation Analysis and discuss its business applications.
4. Compare Correlation and Regression Analysis.
5. Explain the procedure of Hypothesis Testing with a practical example.
6. Analyze the role of Probability Distributions in Data Analytics.
7. Compare Z-test, t-test, Chi-square test, and ANOVA.

Execution (15 Marks)

Experiment: Supply Chain Performance Analysis Using Excel

Part A – Data Preparation

- Import DataCo Smart Supply Chain Dataset.
- Format the dataset as an Excel Table.
- Check for duplicate Order IDs and remove them if any.
- Apply appropriate data formats (Date, Currency, Number).
- Sort the dataset based on Order Date in ascending order.

Part B – Formula-Based Analysis

- Calculate the Purchase Cost for each order.
  - Formula: Quantity Ordered × Unit Cost
- Calculate the Sales Revenue for each order.
  - Formula: Quantity Sold × Selling Price
- Calculate the Gross Profit for each order.
  - Formula: Sales Revenue – Purchase Cost – Shipping Cost
- Determine the Inventory Status using the following criteria:
  - Inventory < 20 → Low Stock
  - Inventory ≥ 20 → Sufficient Stock
- Determine the Delivery Status:
  - Delivery Days ≤ 3 → On Time
  - Delivery Days > 3 → Delayed
- Calculate the Profit Margin (%) for each order.
  - Formula: (Gross Profit / Sales Revenue) × 100

Part C – Descriptive Analytics

- Calculate the total Purchase Cost of all orders.
- Calculate the total Sales Revenue.
- Calculate the total Gross Profit.
- Find the average Delivery Days.
- Identify the order with the highest Gross Profit.
- Find the supplier with the highest average customer rating.
- Count the number of delayed deliveries.
- Determine the transportation mode with the highest shipping cost.

Part D – Pivot Table Analysis

- Create the following Pivot Tables:
- Supplier-wise Purchase Cost.
- Region-wise Sales Revenue.
- Product Category-wise Gross Profit.
- Warehouse-wise Inventory Available.
- Transportation Mode-wise Shipping Cost.
- Supplier-wise Average Customer Rating.

Part E – Data Visualization

- Create the following charts:
- Column Chart showing Product Category vs Sales Revenue.
- Pie Chart showing Transportation Mode distribution.
- Bar Chart showing Supplier-wise Gross Profit.
- Line Chart showing Monthly Sales Revenue.
- Scatter Plot showing Delivery Days vs Customer Rating.

Part F – Dashboard

- Develop an interactive dashboard containing:
- Total Purchase Cost
- Total Sales Revenue
- Total Gross Profit
- Average Delivery Days
- Highest Revenue Product Category
- Best Performing Supplier
- Region-wise Sales
- Inventory Status
- Customer Rating Analysis
- Include Slicers for:
- Supplier
- Region
- Warehouse
- Product Category
- Transportation Mode

Analytical Questions

- Which supplier contributes the highest gross profit? Support your answer with
- appropriate calculations.
- Which transportation mode is the most cost-effective? Justify using shipping cost analysis.
- Which region generates the highest sales revenue? Explain with evidence.
- Which products should be reordered immediately? Justify using inventory status.
- Is there a relationship between delivery time and customer rating? Explain using the scatter plot.
- Which warehouse has the highest inventory level?
- Identify products with low profit margins and suggest corrective measures.
- Recommend three strategies to improve the overall supply chain performance based on your analysis.